

Eightfold-Candidate-Transformer

Candidate Profile Transformer – Technical Design

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Objective

The Candidate Profile Transformer consolidates candidate information from multiple structured and unstructured sources into a single canonical candidate profile. The system resolves conflicts, normalizes data, tracks provenance, and calculates confidence scores while supporting configurable output.

Architecture

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CSV / JSON / TXT

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Loaders

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Normalization

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Merge Engine

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Provenance Tracking

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Confidence Scoring

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Validation

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Projection Engine

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JSON Output

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Components

Loaders

Responsible for reading candidate information from:

- Recruiter CSV
- ATS JSON
- GitHub JSON
- LinkedIn JSON
- Notes TXT

Normalizer

Normalizes:

- Name
- Email
- Phone
- Skills
- Country

into a common format.

Merge Engine

Merges records using source priority.

Priority:

1. Recruiter
2. ATS
3. LinkedIn
4. GitHub
5. Notes

Rules:

- Higher priority wins.
- Empty values ignored.
- Skills merged as union.

- Duplicate values removed.

Provenance

Each merged field stores:

- Source
- Merge Method

This provides traceability.

Confidence

Each source contributes a confidence score.

Source	Confidence
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Recruiter	1.00
ATS	0.95
LinkedIn	0.90
GitHub	0.85
Notes	0.60

Overall confidence is calculated as the average of contributing fields.

Validation

Checks:

- Candidate Name
- Email
- Phone
- Confidence Score

Missing fields generate warnings instead of terminating execution.

Assumptions

- One candidate is processed at a time.
- Recruiter data is most trusted.
- Notes provide supplemental information.
- Country codes follow ISO-3166 Alpha-2.
- Phone numbers follow E164 format.

Future Improvements

- REST API
- Database persistence
- Better NLP for notes
- Multiple candidate support
- UI Dashboard